CSS MEDIA QUERIES

Week 0 - Day 4 | Complete Revision Cheat Sheet

Roman Urdu / English guide. Rule: @media condition check karti hai; condition true hone par andar wali CSS apply hoti hai.

1. Basic Syntax

```
@media (min-width: 768px) {
  .card { background: blue; }
}
```

Meaning: Jab viewport 768px ya us se bari ho, .card par blue background apply karo.

2. Width, Height & Aspect Ratio

Condition	Kab true hogi	Use
min-width: 768px	Width >= 768px	Tablet aur larger
max-width: 767px	Width <= 767px	Mobile
width: 768px	Exactly 768px	Rare exact match
width >= 768px	Modern syntax, >= 768px	Newer range syntax
768px <= width < 1024px	768px included, 1024px excluded	Tablet range
min-height: 600px	Height >= 600px	Tall viewport
max-height: 500px	Height <= 500px	Short viewport
min-aspect-ratio: 16/9	Viewport ratio at least 16:9	Wide screens
max-aspect-ratio: 3/4	Ratio at most 3:4	Tall/narrow screens

3. Orientation and Inputs

Condition	Working
orientation: portrait	Viewport height >= width
orientation: landscape	Viewport width > height
hover: hover	Primary input hover supports
hover: none	Primary input cannot hover
any-hover: hover	At least one input supports hover
pointer: fine	Primary pointer precise (often mouse)
pointer: coarse	Primary pointer less precise (often touch)
pointer: none	No primary pointing device
any-pointer: fine	At least one precise pointer

4. Preferences & Accessibility

Condition	Meaning / CSS use
prefers-color-scheme: dark	Dark scheme preferred; switch colors
prefers-color-scheme: light	Light scheme preferred
prefers-reduced-motion: reduce	Reduce animation/transitions
prefers-reduced-motion: no-preference	No reduced-motion request
prefers-contrast: more / less / custom	Adapt contrast where supported
forced-colors: active	System forces color palette
prefers-reduced-transparency: reduce	Avoid transparency where supported
prefers-reduced-data: reduce	Lightweight assets where supported

Note: Kuch advanced preference queries ka browser support limited ho sakta hai.

5. Media Types & Logical Operators

Syntax	Meaning	Example
screen	Display screens	@media screen {...}
print	Print / print preview	@media print {...}
all	All media types	@media all {...}
speech	Speech-oriented output; limited support	@media speech {...}
and	Both conditions true	(min-width: 768px) and (orientation: landscape)
,	OR: any condition true	(max-width: 480px), (min-width: 1200px)
not	Reverse condition	not (hover: hover)
only	Legacy parser guard	only screen and (min-width: 768px)

6. Advanced / Special Features

Media Feature	Kya check hota hai
resolution / min-resolution	Screen pixel density
color-gamut: p3	P3 color gamut supported
color: 0	No color output
monochrome	Monochrome capability
inverted-colors: inverted	Inverted output colors
display-mode: standalone / fullscreen	App or PWA display mode
scripting: enabled	JavaScript available
update: slow	Slow output refresh

Media Feature	Kya check hota hai
overflow-block: paged	Paginated block overflow
overflow-inline: scroll	Inline scrolling available
dynamic-range: high	HDR display capability
video-dynamic-range: high	HDR video capability
horizontal-viewport-segments: 2	Two horizontal viewport segments
vertical-viewport-segments: 2	Two vertical viewport segments
device-posture: folded	Folded device posture
color-index	Color lookup-table capability
grid	Grid vs bitmap output

Some are experimental / uncommon. Avoid deprecated device-width, device-height, and device-aspect-ratio.

7. W0D4 Task 1 - Three Card Layout

```
.cards { display: grid; grid-template-columns: 1fr; gap: 20px; }
@media (min-width: 768px) {
  .cards { grid-template-columns: repeat(2, 1fr); }
}
@media (min-width: 1024px) {
  .cards { grid-template-columns: repeat(3, 1fr); }
}
```

Viewport	Result	Why
375px	1 column	Default mobile-first CSS
768px	2 columns	768 query true
900px	2 columns	768 true, 1024 false
1024px	3 columns	Both true; later 1024 rule wins
1440px	3 columns	Desktop rule true

8. Other CSS Tools Used on Day 4

CSS	Use
max-width: 100%	Images parent width se bahar na niklein
clamp(24px, 4vw, 48px)	Fluid headings with minimum/maximum
rem / em / vw / vh	Relative sizing
flex-wrap: wrap	Cards next line par shift
gap: 16px	Spacing between items
transition: transform .3s	Smooth visual effect

CSS	Use
:hover	Pointer hover state
:root and var(--name)	Shared color/theme variables

9. Quick Memory Rules

WIDTH = min / max / range; **SHAPE** = portrait / landscape; **INPUT** = hover / pointer; **PREFERENCE** = dark / motion / contrast; **OUTPUT** = screen / print / resolution.

Standup line: I created a three-card grid, began with one mobile column, and added min-width breakpoints at 768px and 1024px to switch to two and three columns. I also practiced viewport testing, mobile navigation, responsive images, dark mode, and transitions.

Reference: developer.mozilla.org/en-US/docs/Web/CSS/Reference/At-rules/@media